Node Lifecycle Hypothesis
and the Crystallisation of 3D Space in CRF

Two-Phase Dynamics, IDV–Ω Duality,
and Five Testable Sub-Hypotheses

Grounding the marble intuition in the δ-chain

Hypothesis Layer — consistent with CRF axioms, pending formal derivation

Ougit Jarittum

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Abstract

CRF Hybrid V18 confirmed that Internal Dimensional Volume (IDV) and the crystallised mass field Ω are anti-correlated across node history: as $\Omega\uparrow$, IDV$\downarrow$. This paper formalises that observation into the Node Lifecycle Hypothesis — a two-phase model of how any node in the δ-chain transitions from informational youth (high entropy, high dimensionality) to matter (low entropy, 3D-locked, gravitationally active).

The core prediction is:

$$\frac{d}{dt}\mathrm{IDV}(t) \approx -\frac{d}{dt}\Omega(t) \qquad \text{at } d_s \approx 3$$

This relation, if verified in V20, would constitute the first direct measurement of the IDV-to-Ω transfer rate — the moment when "possible" becomes "actual."

Five sub-hypotheses (HN-01 through HN-05) extend the core model: phase-transition thermodynamics, node aging and rejuvenation, multi-node resonance, dimensional hysteresis, and causal geometry mapping. Each is stated in CRF-native language with a grounding in existing derivations and a specific V20 test signature.

Contents

1 The Core Hypothesis: Two-Phase Node Lifecycle

Hypothesis 1 (Node Lifecycle)**.** Every node in the δ-chain passes through two phases, separated by a δ-Spark transition:

Phase I — Informational Youth (IDV-dominant):

- $H(P_i)$ high: many patterns accessible
- $\Omega_i = D_{\mathrm{KL}}(C_i \| P_{\mathrm{bg}})$ low: little accumulated history
- d_s fluctuating: node has not yet selected a stable geometry
- IDV_i high: large internal dimensional volume

Transition — δ-Spark: D_{KL} gradient exceeds $\theta_i = 1 - e^{-D_{\mathrm{KL}}/D_0}$, triggering an R-event cluster. The SO(3) attractor (proved in CRF_SO3_Proof) selects $n = 3$.

Phase II — Informational Maturity (Ω-dominant):

- $H(P_i) \to \varepsilon_0$: constraint distribution nearly deterministic
- Ω_i high: deep R-event history, gravitationally active
- $d_s \to 3$: geometry locked by SO(3) symmetry
- IDV_i low: internal space has crystallised into 3D

Grounding in Prior Results

The following are already derived and support the core hypothesis:

- $\Omega_i = D_{\mathrm{KL}}(C_i \| P_{\mathrm{bg}}) = \mathrm{mass}_i - 1$ (EIC Bridge central identification)
- $H(t) = H_0/(1+\alpha H_0 t)$ approaches 0 asymptotically (CRF Entropy Dynamics)
- $n = 3$ is the unique δ-stable dimension (CRF SO(3) Proof, Theorem 1)
- IDV $\uparrow \Leftrightarrow \Omega \downarrow$ confirmed empirically (V18 simulation)
- Gravity $R^2 = 0.908$ emerges from Ω-phase nodes (V18)

1.1 Core Prediction

Central Testable Prediction

At the moment of δ-Spark (when d_s crosses 3.0):

$$\frac{d}{dt}\mathrm{IDV}_i(t) \approx -\frac{d}{dt}\Omega_i(t)$$

The rate at which internal dimensional volume is lost equals the rate at which crystallised mass is gained. This is the IDV-to-Ω *transfer rate* — the informational cost of becoming actual.

V20 test: measure $\mathrm{IDV}(t)$ and $\Omega(t)$ simultaneously at per-node resolution during a collapse event. Signature: the two time derivatives are equal and opposite at $d_s \approx 3.0$, within simulation noise.

2 HN-01: Phase Transition Thermodynamics

Hypothesis 2 (Phase Transition Thermodynamics)**.** The IDV-to-Ω transition can be characterised by a *free informational ratio*:

$$F_i(t) = \frac{H(P_i)}{\Omega_i + \varepsilon_0}$$

Phase I: F_i high (H large, Ω small). Phase II: F_i low (H small, Ω large). The δ-Spark occurs when $dF_i/dt < 0$ passes a critical rate.

Why this formulation, not $F = H - \Omega$: H has units of nats (information) and Ω has units of D_{KL} (dimensionless divergence). Subtraction is not dimensionally consistent. The ratio $H/(\Omega + \varepsilon_0)$ is dimensionless and correctly captures the balance: when $\Omega \gg H$, $F \to 0$ (matter); when $H \gg \Omega$, $F \to \infty$ (field).

Grounding

$H \to 0$ from CRF Entropy Dynamics: $dH/dt = -\alpha H^2$. $\Omega = D_{\mathrm{KL}}(C \| P_{\mathrm{bg}})$ from EIC Bridge. $\varepsilon_0 = |I_{\mathrm{eff}} - \ln 2| = 0.00755$ prevents division by zero and is the minimum instability floor (EIC SINDy confirmed).

V20 Signature

Track $F_i(t)$ per node. Prediction: F_i decreases monotonically during Phase I, drops sharply at δ-Spark, and reaches a floor $F_{\min} \approx \varepsilon_0/\Omega_{\max}$ in Phase II. The collapse is a phase transition in F_i-space, not merely a change in d_s.

3 HN-02: Node Aging and Rejuvenation

Hypothesis 3 (Node Aging and Rejuvenation)**.** An Ω-dominant node does not remain in static Phase II. It *ages*: H continues to decrease toward ε_0. If H falls below ε_0 without renewal, the node loses causal activity (no new R-events) and approaches dissolution.

Rejuvenation occurs when new R-events arrive from the environment (neighbouring high-IDV nodes), restarting the H dynamics. The node transiently returns toward Phase I before re-crystallising.

Grounding

From CRF Entropy Dynamics: $H(t) = H_0/(1 + \alpha H_0 t) \to 0$ asymptotically. This is valid for an *isolated* node. A non-isolated node receives D_{KL} perturbations from neighbours, which restart the H-decay clock.
In physical terms: an isolated crystallised node ages (approaches the Schwarzschild-like entropy boundary from HDynamics). A node in contact with an active environment is refreshed. This is the CRF account of matter interacting with radiation.

V20 Signature

Monitor entropy $H_i(t)$ for Ω-dominant nodes. Prediction: H_i decays toward ε_0, not to zero. At $H_i = \varepsilon_0$: if the node is isolated, Ω begins to decrease (aging); if connected to active neighbours, H_i is refreshed (rejuvenation). Observable: anti-correlated oscillations of H_i and Ω_i in long-running simulations.

4 HN-03: Multi-node Resonance

Hypothesis 4 (Multi-node Resonance)**.** When N_c nodes undergo δ-Spark near-simultaneously, their collective Ω-field exceeds the sum of individual contributions:

$$\Omega_{\text{collective}}(t) > \sum_{i=1}^{N_c} \Omega_i(t)$$

The excess arises from constructive interference of the wave fields Φ_i, which creates an additional R-event bonus.

Grounding — Strongest of the Five HNs

CRF Synchronicity (derived): $\Pr(\text{sync}) \propto e^{-\alpha D_{\text{KL}}}$. When pairwise $D_{\text{KL}} \to 0$ (Axiom 11 direction), R-events synchronise.
V18 already measures this as `local_info_weight`:

$$\text{interference}_i = \sum_{j \in \mathcal{N}(i)} \max(0, \cos(\Phi_j - \Phi_i)) \cdot D_{\text{KL}}(P_i \| P_j)$$

`local_info_weight` accumulates this bonus. $\Omega_{\text{collective}} > \sum \Omega_i$ if and only if $\sum_i \text{interference}_i > 0$ — which V18 confirms is positive.
This is the CRF mechanism for gravitational well formation: matter clusters because synchronised collapse creates more Ω than isolated collapse.

V20 Signature

Inject a synchronised δ-Spark into a cluster of N_c nodes. Compare $\Omega_{\text{collective}}(t)$ against $\sum_i \Omega_i(t)$. Prediction: the collective exceeds the sum by a factor proportional to $e^{-\overline{D_{\text{KL}}}}$ of the cluster.

5 HN-04: Dimensional Hysteresis

Hypothesis 5 (Dimensional Hysteresis)**.** The transition IDV-phase $\to$ Ω-phase is irreversible. A perturbation applied to a node in Phase II (3D-locked, high Ω) cannot restore the node to Phase I (high-D, low Ω) as a stable state. The node may oscillate near $d_s = 3$ under perturbation, but returns to Phase II when the perturbation ceases.

This follows directly from SO(3) Proof

The SO(3) Proof (Theorem 1) established that $n = 3$ is the *unique* dimension satisfying both δ-stability conditions: non-abelian (Axiom 0) and rank-1 (Axiom 11).
Corollary: there is no stable high-D state. A perturbation can temporarily increase d_s above 3, but the system has no stable attractor at $d_s > 3$. The perturbation decays and d_s returns to 3.
This is hysteresis: the forward path (high-D $\to$ 3D) passes through a δ-Spark; the backward path (3D $\to$ high-D stable) does not exist. The loop is not closed — a defining property of hysteresis.
Status: **Corollary** of SO(3) Proof, not merely hypothesis.

V20 Signature

Apply entropy injection (noise) to a crystallised 3D-locked node. Measure $d_s(t)$ during and after perturbation. Prediction: d_s rises transiently, then returns to ≈ 3. The hysteresis loop $\oint d_s \, dt \neq 0$ is measurable as the area of the d_s-vs-time curve during perturbation and recovery.

6 HN-05: Causal Geometry Mapping

Hypothesis 6 (Causal Geometry Mapping)**.** The causal connectivity of node i is its accumulated R-event count $N_{\text{eff},i} = \texttt{local\_time}[i]$. Ω-dominant nodes have higher N_{eff} and therefore higher causal connectivity than IDV-dominant nodes. The local metric tensor $G^{(i)}$ is the geometric expression of this connectivity: it encodes *which directions* the node's R-event history has reinforced.

Grounding

From CRF core: $t = |R_{\text{events}}|$ and $\text{mass}_i = 1 + \kappa \cdot \texttt{local\_time}[i]$. Therefore $\Omega_i = \text{mass}_i - 1 = \kappa \cdot N_{\text{eff},i}$. Higher Ω = more R-events = more causal connections.
The metric tensor $G^{(i)}$ is updated at each R-event (V17/V18 simulator):

$$G^{(i)} \mathrel{+}= \lambda \cdot \text{IDV}_i \cdot \cos(\Phi_j - \Phi_i) \cdot \mathbf{r}_{ij}^{(3)} \otimes \mathbf{r}_{ij}^{(3)}$$

Each update reinforces the 3D directions of the R-event. After many R-events, $G^{(i)}$ has strong 3D eigenvalues — the node's causal geometry has crystallised.
Causal connectivity: $C_i = \text{tr}(G^{(i)})$, the total geometric weight of the node's accumulated history.

V20 Signature

Compare $C_i = \text{tr}(G^{(i)})$ for IDV-dominant and Ω-dominant nodes. Prediction: $C(\Omega) \gg C(\text{IDV})$, with the jump in C_i coinciding with the δ-Spark collapse event.

7 V20 Experiment Design

What V20 Must Measure

All five hypotheses have predictions that can be tested in a single simulation run if V20 adds the following detectors:

1. **IDV-to-Ω transfer rate:** $\dot{\mathrm{IDV}}_i$ and $\dot{\Omega}_i$ per node. Core prediction: equal and opposite at $d_s \approx 3$.
2. **Free informational ratio:** $F_i = H(P_i)/(\Omega_i + \varepsilon_0)$ per node per sweep. HN-01: F_i drops sharply at δ-Spark.
3. **Entropy floor tracking:** Flag nodes where $H_i < 2\varepsilon_0$. HN-02: monitor for aging/rejuvenation cycles.
4. **Collective Ω excess:** Compare Ω_{cluster} against $\sum_i \Omega_i$ for synchronised collapse clusters. HN-03: excess > 0.
5. **Hysteresis test:** Inject entropy into 3D-locked nodes and measure d_s recovery. HN-04: d_s returns to 3, not to 8.
6. **Causal connectivity:** $C_i = \mathrm{tr}(G^{(i)})$ per node. HN-05: jump at collapse, $C(\Omega) \gg C(\mathrm{IDV})$.

8 Complete Status Table

Code	Title	Status	Grounding
Core	Node Lifecycle (two-phase)	Hypothesis	V18 IDV–Ω anti-corr.
HN-01	Phase Transition Thermo.	Hypothesis	HDynamics + EIC
HN-02	Node Aging & Rejuvenation	Hypothesis	ε_0 floor + HDynamics
HN-03	Multi-node Resonance	Hypothesis	CRF Synchronicity (derived)
HN-04	Dimensional Hysteresis	**Corollary**	SO(3) Proof (derived)
HN-05	Causal Geometry Mapping	Hypothesis	$t = \lvert R_{\text{events}} \rvert$ + V18 metric

All hypotheses are consistent with the δ-chain. None requires a new axiom.

The marble does not choose to crystallise.
It simply runs out of reasons not to.
When possibility has been fully explored,
only actuality remains.